

Survival analysis: Modeling time-to-event outcomes

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15th September, 2026



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What we will cover....

- 1 Introduction
 - Issues with survival data
 - Survival distributions and some important functions
- 2 Bivariate analysis of survival data
 - Kaplan-Meier survival curves
 - Descriptive statistics
 - Comparing groups
- 3 Modeling survival data
 - Introduction to Cox PH regression
 - Example: Cox PH regression
 - Assumptions of Cox PH regression
 - Other survival analysis models

Time-to-event data

Often we are interested in an outcome involving time to an event. For example in clinical research:

- Time to death
- Time to remission (after treatment)
- Length of stay (e.g. in ICU)
- Time to recidivism (after addiction treatment)

All of these endpoints come under the heading of *survival data* (Even though not all involve death as an endpoint).

Most survival analysis are performed in the clinical setting.

Only other **common** application is to the survival of products.
e.g. Modeling time to computer failure → pricing warranties

Issues with survival data

Analysing survival data presents a few problems not encountered in other types of analysis. Specifically:

- Data truncation and censoring
- Conditional probability: To consider the probability that a subject might die at particular point in time, we have to account for the fact that they have survived to that point i.e. The probability of subject X dying on day 30 is conditional on patient being alive on all preceding days
- A survival outcome has **TWO** components: the continuous variable, **Time**; AND the binary variable of whether the event happened or not, **Failure** (e.g death status)

Types of censoring and truncation

Examples of censoring and truncation

Subjects that do not experience event of interest (B,C,D,E)

Incomplete follow-up

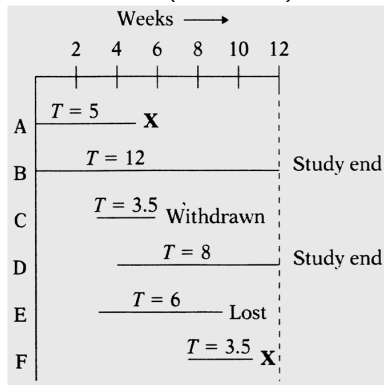
- Lost to follow-up (E)
- Withdraws from study (C)
- Dies of something other than clinical end point -if *Disease-free survival is endpoint*

Right censored (B,C,D and E)

- B and D - study ends
- C and E - withdrew/LTFU

Left truncated (A,B)

- A and B - time at risk prior to study



The event of interest (e.g. death from MI) only happens in two patients (A and F) during follow up

Distribution of survival times

First, consider the Probability Distribution (density function), $f(t)$, and Cumulative Probability Distribution, $F(t)$ of survival times, t :

$f(t)$ represents the probability of dying on a particular day, t

$F(t)$ represents the probability of dying any time up to a particular day, t

Example:

$f(23)$ = probability of dying day 23; and

$F(23)$ = probability of any time in the first 23 days

However, generally we are more interested in the probability that someone will **survive** up to a particular point in time, which brings us to the survival function, $S(t)$

Some important functions

Survivor function, $S(t)$ and the hazard function $h(t)$

Rather than $f(t)$ and $F(t)$, focus in survival analysis is on the **survival** and **hazard** functions

- **Survival function**, $S(t)$, is the probability of surviving longer than time, t **e.g. Prob(survival time > 1 yr)**

$$S(t) = P(T > t) = \int_t^{\infty} f(u) du = 1 - F(t)$$

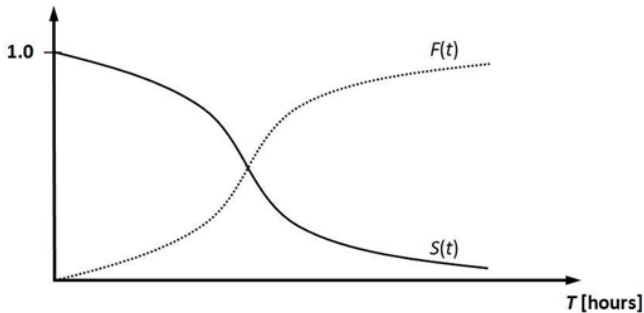
- **Hazard function**, $h(t)$, **instantaneous** risk of event at time, t

$$h(t) = \frac{f(t)}{S(t)}$$

Hazard function: the probability that if you survive to t , you will die in the next instant

$F(t)$ Vs $S(t)$

Relationship between $F(t)$ and $S(t)$: $S(t) = 1 - F(t)$



Analysis of survival data

Like all of the other types of analysis, survival analysis can be used to generate:

- ① Descriptive statistics (including graphs)
- ② Bivariate methods (Specifically: Survival in terms of a single categorical covariate)
- ③ Multivariable models e.g. Proportional Hazards [*aka* Cox] regression

The Kaplan-Meier survival curve

- Perhaps the most common graphical representation of survival data
- Represents an **estimator** of the (true) surv.curve, $S(t)$
- *aka* the product-limit formula (see next slide)
- Purely empirical and non-parametric (makes no 'distributional' assumption about survival times)
- Accounts for censoring
- Generates the characteristic 'stair step' survival curves
- Does not account for confounding or effect modification by other covariates
- Contrary to popular belief, KM curves are not a 'test', they are just a graph

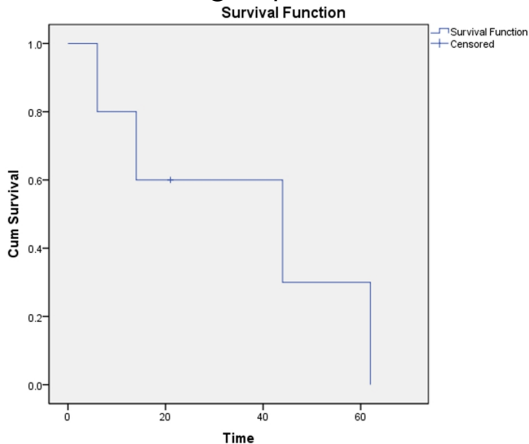
Simple example of a survival curve

<i>Subject</i>	<i>Survivaltime</i>	<i>Died(1)/Censored(0)</i>
1	6	1
2	44	1
3	21	0
4	14	1
5	62	1

<i>Subject</i>	<i>calculation</i>	<i>S(t)</i>
$0 \leq t < 6$	$\frac{5}{5}$	1.0
$6 \leq t < 14$	$1.0 \times \frac{4}{5}$	0.8
$14 \leq t < 21$	$1.0 \times \frac{4}{5} \times \frac{3}{4}$	0.6
$21 \leq t < 44$	$1.0 \times \frac{4}{5} \times \frac{3}{4} \times \frac{3}{3}$	0.6
$44 \leq t < 62$	$1.0 \times \frac{4}{5} \times \frac{3}{4} \times \frac{3}{3} \times \frac{1}{2}$	0.3
$t > 62$	$1.0 \times \frac{4}{5} \times \frac{3}{4} \times \frac{3}{3} \times \frac{1}{2} \times \frac{0}{1}$	0

KM curve for simple example

and the resulting Kaplan-Meier curve....



Motivating example: Worcester Heart Attack Study

Background

- Before going into more detail about survival analysis, let's consider a real data set.
- Study considering factors and time trends associated with long-term survival following a myocardial infarction among residents of Worcester, Massachusetts
- We'll consider a trimmed down version of the data with a small selection of covariates (continuous and categorical) and a small subset ($n=500$) of patients from the full study.

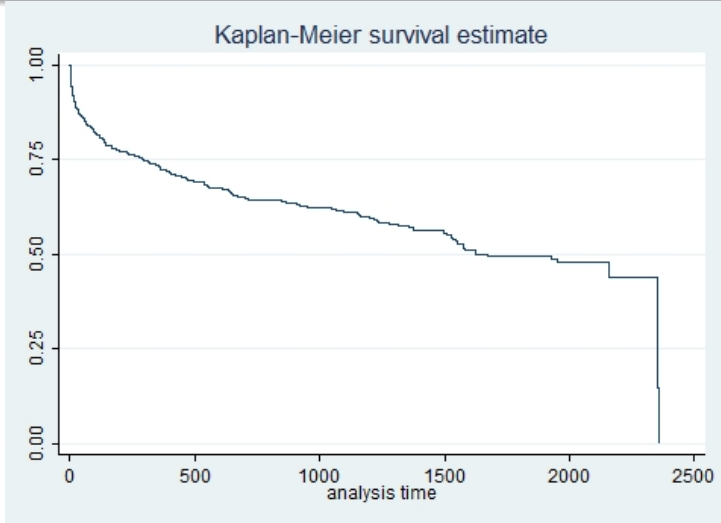
Motivating example: Worcester Heart Attack Study

Variables in dataset

Covariates	Covariates (cont.)
Id	MI type (0=non-Q wave, 1=Q wave)
Age	Cohort year (1=1997, 2=1999, 3=2001)
Gender (0=male, 1=female)	Time and censoring variables
Initial heart rate	Admission date (a)
SBP	Discharge date (b)
DBP	Last follow-up date (c)
BMI	Length of stay (b-a)*
Family history of CVD(0=n,1=y)	Discharge status(0=alive, 1=dead)#
Atrial fibrillation(0=n,1=y)	Total length follow-up (c-a)*
Cardiogenic shock(0=n,1=y)	Status at last follow-up (0=alive, 1 dead)#
Congest. Heart compl(0=n,1=y)	
Complete heart block(0=n,1=y)	# censoring variables *analysis-time variables
MI order (0=first,1=recurrent)	

KM curve for Worcester data

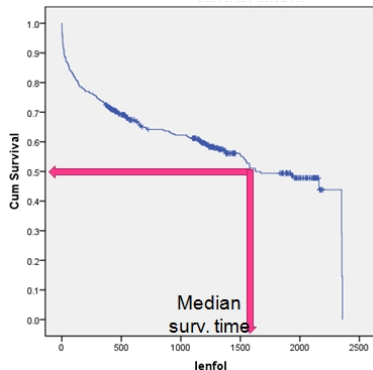
All subjects for Total follow-up



Descriptive statistic: Median survival time

Using the KM curve

- Often clinicians want to know what is the survival time that 50% of individuals can expect to survive (Median survival time)?
- Software can calculate the median survival time, or we can just read it off the KM-curve.
- It is the time associated with the 0.5 point in cumulative survival



Descriptive statistic

Mean vs. median survival time

- As survival times are zero bound on the left (can't have survival < 0) we would expect survival times to be positively skewed.
- So, the median rather than mean survival time is the best way to represent the 'typical' survival time.

Statistic	Survival
records	500.00000
events	215.00000
rmean	1417.21461
se(rmean)	48.02453
median	1627.00000
0.95LCL	1527.00000
0.95UCL	NA

These values are routinely output with KM curves

Descriptive statistic

All the Quartiles

We can also obtain all the quartiles

Table: Summary statistics from survival curve

Quartile	Survival
25	296
50	1627
75	2353

- 75% of individuals survived at least 296 days,
- 50% survived at least 1627 days (median survival time)
- 25% survived at least 2353 days

Comparing groups

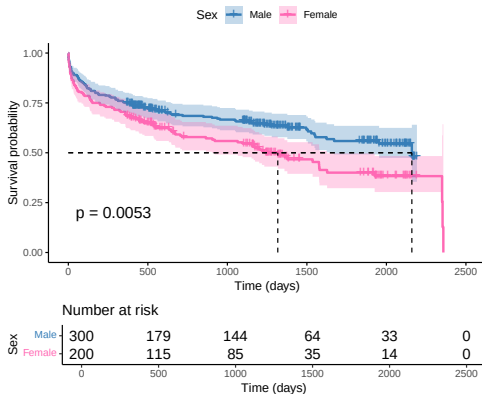
Methods

- Visualizing the entire dataset well and good, but unless our sample represents a single homogeneous population, we need methods to compare groups of individuals
- A number of tests exist able to compare the survival curves arising from different groups (e.g. Survival time for those on treatment A vs. those on treatment B).
- The **bivariate** test we consider here is non-parametric (makes no assumption about the distribution of survival times from each group).
- This method is called the **log-rank test**.

Comparing groups: Heart attack data

Comparing KM curves

EYEBALLS: First, we can **visualize** group differences by generating group-specific KM curves



Males **APPEAR TO** have a more favorable survival experience

Comparing groups: Heart attack data

Bivariate hypothesis testing

Now let's look at the descriptive statistics (median survival time) for each group

Percentiles

gender	25.0%		50.0%		75.0%	
	Estimate	Std. Error	Estimate	Std. Error	Estimate	Std. Error
0	.	.	2160.000	.	354.000	109.695
1	2353.000	176.168	1317.000	177.039	151.000	84.201
Overall	2353.000	79.465	1627.000	159.555	295.000	63.223

- The median survival time for males (0) was 2160 days and for females (1) 1317 days.
- **Questions:**
 - ① Why is there no estimate (or standard error) for the the 25% survival time of males?
 - ② Why is there no standard error for the median survival time for males?

Comparing groups: Heart attack data

Bivariate hypothesis testing

Now let's see if there is a (statistically) significant difference in the expected survival of males compared to females

```
> survdiff(wHAS500.surv ~ gender, data = df, rho=0)
Call:
survdiff(formula = wHAS500.surv ~ gender, data = df, rho = 0)

              N observed Expected (O-E)^2/E (O-E)^2/V
gender=0 300      111    130.7      2.98      7.79
gender=1 200      104     84.3      4.62      7.79

chisq= 7.8 on 1 degrees of freedom, p= 0.005
```

- Log-rank demonstrates a difference in the survival experience between the groups ($\chi^2_{LR}=7.8$, $p<0.01$)

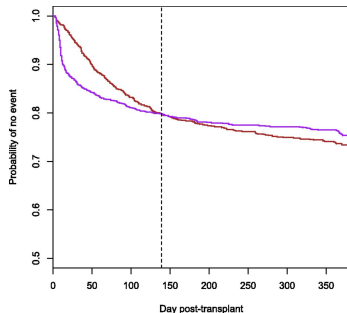
Problem with Log-rank tests

Log-rank tests (and other classical bivariate tests) are YES/NO answers. They don't tell us much about the nature (or magnitude) of the association. We need better methods

Comparing groups

A final word on KM curves

- In order to conclude a difference in the survival of two or more groups, **their survival curves should not intersect** (Regardless of Log rank test's results).
- Here, patients on **Treatment 1** appear to have a better prognosis in the first 140 days, but after that those in the **Treatment 2 group** fare better.



This is a case of **severe disproportionately** (more later)

Where to from here?

Modeling survival with Cox Proportional Hazards Regression

- Now we will consider a true **model** of survival times.
- Specifically we will consider a model that can deal with multiple categorical and/or continuous predictor: **Cox proportional hazards regression**
- You will see that Cox regression is a MUCH more useful method than those we have covered so far, but it makes a few restrictive assumptions too

Cox proportional hazards regression

- Most widely used methods for modelling time-to-event (survival) data
- Cox PH regression is **semi-parametric**; it makes no assumptions about the form of the survival distribution (or more specifically, the hazard distribution)
- However, nor is PH regression fully nonparametric as it makes assumptions about the functional form of the covariates (i.e. it is model based), and the central assumption of **proportionality** (more later)

The Cox proportional hazards model

$$h_i(t) = h_0(t)e^{\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k}$$

$$h_i(t) = h_0(t)e^{X\beta}$$

where

$h_i(t)$ is hazard for subject i at time t

h_0 is the baseline hazard (common to all subjects)

$X\beta$ is the subject-specific component; and

the natural base exponential function, e , as the inverse link function (i.e. a log link) implying a log-linear relationship between the hazard and the covariates

The proportional hazards model

$$h_i(t) = h_0(t)e^{X\beta}$$

- The covariates (in this form of the model) are time-independent: often called **baseline covariates**
 - i.e. there is no t component in the $X\beta$.
- Baseline covariates are typically represented by variables such as age, sex etc. and there are assumed **NOT** to vary throughout the survival experience
- To consider time-varying covariates (e.g. SBP and DBP) we need to extend the PH model to include β that are dependant on t , i.e. $\beta(t)$ (we won't do this today)

Hazard ratios

- The parameters estimated in Cox PH regression, β , can be transformed to obtain are hazard ratios.
- As with Logistic regression (ORs) and Poisson Regression (RRs), $\hat{\beta}$, can be **exponentiated** to get the hazard ratio (HRs). That is, $HR = e^{\beta}$
- HRs represent the risk of death (or the clinical endpoint) for one group relative to that of another (categorical covariates), or change in hazard as we increase a unit (continuous covariates).

Example: consider a single categorical covariate with two classes (Gender). If we dummy code Gender = 0 for males and Gender = 1 for females. Then...

$$h_i(t) = h_0(t)e^{\beta \text{Gender}}$$

Hazard ratios: A measure of effect size

If we performed a Cox PH regression and found:

$$\beta_{\text{Gender}} = 2$$

Then:

$$h_{\text{males}}(t) = h_0(t)e^{2 \times (0)} = h_0(t)$$

$$h_{\text{females}}(t) = h_0(t)e^{2 \times (1)} = h_0(t)e^2$$

$$HR = \frac{h_0(t)e^2}{h_0(t)} = e^2 = 7.4$$

This means: *Mortality in Females is 7.4 time more likely(assuming death represents the hazard) compared to that in males*

A clever little trick is being used here. Can you spot it?

Recall our Worcester Heart Attack Study (n=500)

Covariates	Covariates (cont.)
Id	MI type (0=non-Q wave, 1=Q wave)
Age	Cohort year (1=1997, 2=1999, 3=2001)
Gender (0=male, 1=female)	Time and censoring variables
Initial heart rate	Admission date (a)
SBP	Discharge date (b)
DBP	Last follow-up date (c)
BMI	Length of stay (b-a)*
Family history of CVD(0=n,1=y)	Discharge status(0=alive, 1=dead)#
Atrial fibrillation(0=n,1=y)	Total length follow-up (c-a)*
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Congest. Heart compl(0=n,1=y)	
Complete heart block(0=n,1=y)	# censoring variables *analysis-time variables
MI order (0=first,1=recurrent)	

Modeling effect of MI type and BMI on survival

- First we will consider each covariate separately (i.e. in separate bivariate models) which will give us **unadjusted** (aka crude) effects of the risk factors
- Then both in a single multivariable same model to see:
 - Does taking both covariates into account (simultaneously) improve the model?
 - Does one covariate confound the other? OR RATHER, Does BMI confound MI type (our study effect)?
 - Prognostic/risk factors will then be mutually adjusted→adjusted hazard ratios
- Now our two prognostic factors are:
 - **MI** (0:non-Qwave; 1:Qwave) is binary-**the study effect**
 - **BMI** is continuous and in this case is considered as time-independent (a baseline covariate)

Results:

Frequency table: Recurrent MI

```
> table(df$mitype.fac)
```

Non-Qwave	Qwave
347	153

Frequency table: Infarction type

```
> table(df$mitype.fac)
```

Non-Qwave	Qwave
347	153

- Of the 500 participants, 215 had recurrent heart attacks (so 285 were censored)
- Most participants had Non-Qwave recurrent MIs (note: Non-Qwave is the referent)

Results: Model 1a: (MI type)

	<i>Dependent variable:</i>
	WHAS500.surv
mitype.facQwave	0.517 (0.189, 0.845) t = 3.092 p = 0.002
Observations	500
R ²	0.034
Max. Possible R ²	0.993
Log Likelihood	-1,218.593
Wald Test	15.590*** (df = 1)
LR Test	17.456*** (df = 1)
Score (Logrank) Test	16.144*** (df = 1)

- Significance test of β (Wald-test) gives $p < 0.05$, implying difference in the hazard between the two groups
- Equivalently, the Hazard ratio, e^{β} , is significantly different from 1 (HR=0.517, $p < 0.001$, 95%CI: 0.19-0.85).
- $\Rightarrow$ those with Qwave MI have 0.517 the chance of recurrent relative to non Q-wave MI $\sim$ those with Q wave MI have a 48.3% lower chance of MI (relative to non-Qwave MIs) **at any given time**

Results: Model 1b (BMI)

	<i>Dependent variable:</i>
	WHAS500.surv
bmi	0.906 (0.877, 0.935) t = 61.436 p = 0.000
Observations	500
R ²	0.092
Max. Possible R ²	0.993
Log Likelihood	-1,203.156
Wald Test	44.570*** (df = 1)
LR Test	48.330*** (df = 1)
Score (Logrank) Test	44.307*** (df = 1)
Note:	*p<0.1; **p<0.05; ***p<0.01

- As in other linear models, a unit change in a continuous covariate is often quite small (UPSHOT: Wouldn't expect HR estimates to be as profound)
- $HR = e^{\beta} = 0.906$ (p<0.001; CI:0.88-0.94)
- Interpretation: As we go up one unit of BMI, the chance of death reduces by $(1-0.906)100\% = 9.4\%$.

Now a Multivariable model

- Let's try fitting both of our covariates. We will consider MI type as the study effect (effect of interest) and BMI as a potential confounder.
- We might assume a change in a coefficient (Crude→Adjusted in the effect of interest of 10%) implies confounding.
- Some people insist on only look at change in β s nor OR s, for me I don't think it makes much of a difference (i.e. Change in coefficient $\neq$ Change in HR)

Note: In bivariate model (**Model 1a**) $OR_{MItype} = 0.517$

Multivariable model: Model 2 (MI type and BMI)

- We have >1 covariate (i.e. multivariable model) $\rightarrow$ first test overall model ($\chi^2_{LRT} = 61.86 * ** \Rightarrow$ overall model significant)

<i>Dependent variable:</i>	
WHAS500.surv	
mitype.facQwave	0.555 (0.225, 0.885) t = 3.297 p = 0.001
bmi	0.910 (0.881, 0.939) t = 62.006 p = 0.000
Observations	500
Log Likelihood	-1,196.392
LR Test	61.856*** (df = 2)

- Both covariates are highly significant ($p < 0.001$).
- Compared to $OR_{MItype} = 0.52$ (from model 1a), we now have $OR_{MItype} = 0.56$ (from Model 2)

$$\Rightarrow 100\% \times \frac{0.52 - 0.56}{0.52} = 7.7\% \text{ change (i.e. BMI is not confounder)}$$

Cox regression assumptions

There are two main assumptions associated with Cox regression:

- ① If continuous covariate are associated with the hazard than it is log-linearly related i.e. Is the $\text{Log}(\text{Hazard})$ linearly related to covariates
- ② Hazards remain proportional over whole survival experience
 - This 2nd assumption central to Cox **Proportional Hazards** regression
 - Required to use a Maximum Likelihood like estimator to get the HR (**Or our clever trick doesn't work**)
 - Means HR remains constant (for all survival times).
 - If one group has 2x chance of dying at day 1 relative to the referent, than this holds for day 50, 100, 150 ...

Assessing the proportional hazard assumption

There are two main approaches to assessing the Proportional hazards assumption:

- 1 The log minus log survival plot
- 2 Schoenfeld residuals

Method 1: Log minus log survival plot

This is literally the $\ln(-\ln(S(t)))$ for each level of the covariate (where $S(t)$ is the Kaplan-Meier estimate of the survival curve).

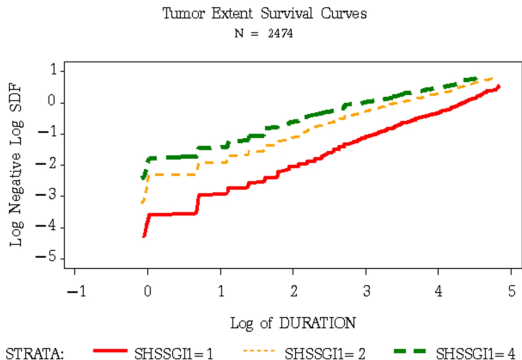
Generally, if the log minus log survival curves are **parallel** for each level of the covariate (e.g. COHORT=1997, 1999 and 2001) then the proportional hazard assumption is met.

Log minus log plots

- As the $\ln(-\ln(S(t)))$ curves seem to be parallel, then the proportional hazards assumption seems to be fine

Problem with Log minus log plots:

- Problem 1: What if we have a continuous covariate?
- Problem 2: Subjective



Not our problem!

This LML plot is from another study. I didn't use the ILML plot in our problem as I used SPSS to conduct the analysis, and SPSS's LML plots are suspicious (at least my version) → wrong direction

Assessing the PH assumption

- The second way of assessing the PH assumption, Schoenfeld residuals, can:
 - ① Consider the PH assumption for continuous covariates
 - ② Provides a (formal) test of the PH assumption
- I am not a big fan of 'tests of assumptions' because they can inadvertently be under- or over-powered in studies which are otherwise meticulously designed.
- However, I consider Schoenfeld residuals an exception to the rule since considering survival curves is complicated (i.e. Not as simple as checking the histogram for normality, for example).

Other multivariable models

- There are various extensions of, and alternatives to, Cox regression to deal with a variety of problems. I will briefly mention a few
- I will not be exhaustive (survival analysis is a whole sub-discipline of biostatistics), but I will theme these advanced models in terms of the problems (be they research question-based or data-based):
 - ① Disproportionality
 - ② Competing risks
 - ③ Recurrent events and time-varying covariates (not equivalent, but related)

Disproportional hazards→difference in baseline survival

- Disproportionality (in plain language) really means there is difference in the baseline survival distribution for different groups of patients.
- Cox regression assumes that ANY and ALL difference in patients survival can be explained **purely** by covariates in the model (beyond this all patients are the same)
- When this is not the case we need to account for the different 'subpopulations' in our sample. Two approaches commonly used:
 - ① **Parametric survival models** - E.g. Exponential and Weibull distribution survival models
 - ② **Stratified Cox regression** - different baseline hazard allowed for different subpopulations. E.g. In an international datasets different populations may have different baseline risk profiles

Stratified Cox regression

If this occurs for a single covariate only (preferably categorical), we can use the Stratified Cox regression model. In this model, a different baseline hazard is allowed for each of stratum, taking the form:

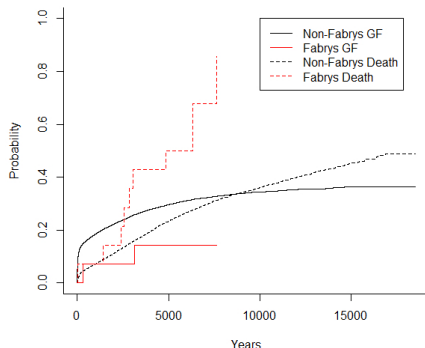
$$h(t|s_i) = h_{0i}(t)e^{\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k}$$

for stratum $i = 1, 2, \dots, m$ strata

- **Note:** While the baseline hazards are allowed to differ for each stratum; the effect of the covariates (on survival) remains the same (i.e. the β s do not vary across strata)
- For that reason, the output from a stratified Cox regression is no more complicated than a standard Cox regression

Competing risk models

- If we want to consider two **independent** endpoints, then we will often go to competing risks analysis
- In the example here we are looking for outcomes in kidney disease patients with or without Fabrys disease (the study effect)
- Here our competing risks (endpoints) are Graft failure (GF) and Death (but not due to graft failure)
- Note that in competing risk analysis the cumulative incidence curve ($F(t)$, the cumulative survival distribution) is used instead of $S(t)$. Recall $S(t) = 1 - F(t)$ and $F(t) = 1 - S(t)$
- Here probability of death appears higher in Fabrys patients, but probability of graft failure appears lower.



Recurrent events and time-varying covariates

While these may seem like very different 'problems' how we deal with them is quite similar. We can see:

- **Recurrence as a repeated value of the outcome.**
E.g. Relapse, childbirth event for a mother etc; and
- **Time-varying covariates as repeated values of the predictors** E.g. SBP measure at baseline, 5 days, 10 days etc

In both of these cases we can use the **Frailty model** (or its friends) to deal with the data clustering that such designs imply

The Frailty model and other related model

- The **Frailty model** is a relatively simple mixed effect model (i.e. contains a patient-specific random effect) and works in a similar way to LMM and GLMM (e.g. linear and generalized linear mixed effect models)
- Cox regression and parametric model versions are available
- How they work is by including a multiplicative random effect, a patient's frailty, to the model and this random effect remains latent - equates to a subject specific baseline hazard
- Extensions to the basic frailty models (e.g. **Cox Mixed Models**) now exist that allow more complex multilevel designs (e.g. Recurrent events AND time-varying covariates, multicentre studies of recurrent events)

Any questions??????

Thank-you!!!!!!
QUESTIONS???